Assessing Student Learning in Computer Science and Technology

In Light of AI Usage

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What best describes your field of study?

Natural Sciences (Biology, Chemistry, Physics)	0%
Social Sciences & Humanities	0%
Health & Medicine	0%
Business, Law & Economics	0%
Formal Sciences (Mathematics, Computer Science)	0%
Applied Sciences (Medicine, Engineering)	0%

What best describes your field of study?

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Applied Sciences (Medicine, Engineering)	##.## %

RESULTS SLIDE

Traditional Assessments in CS

- Written assignments
- Programming Assignments
- Projects
- Exams
- Quizzes

Traditional Assessments in CS (Resilient to AI)

~~Written assignments~~

~~Programming Assignments~~

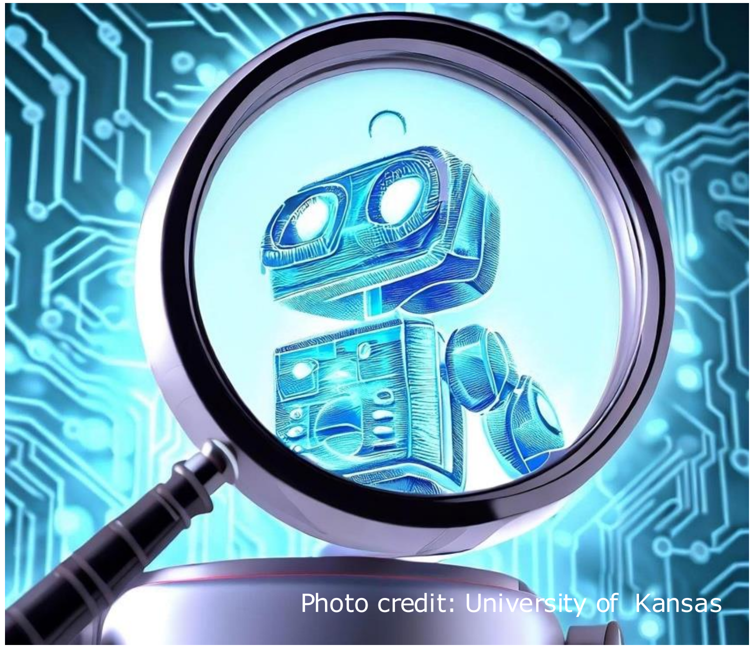
~~Projects~~

Exams

Quizzes

Takeaway: We need a new perspective on assessments in the post-AI era.

Approaches to Consider



Cheating Detection and Prevention

Alternative Assessment Formats

Re-alignment of Course Outcomes

Approaches to Consider

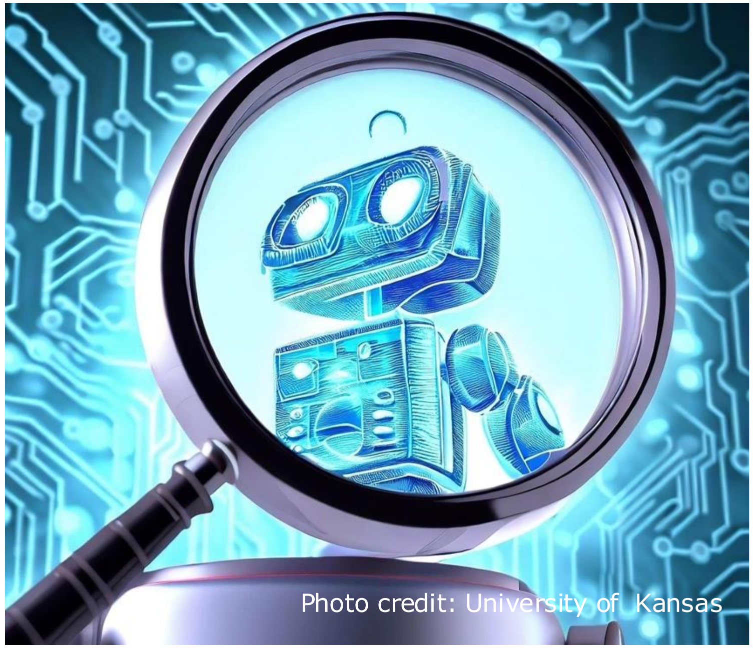


Photo credit: University of Kansas

Cheating Detection and Prevention

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Alternative Assessment Formats

Re-alignment of Course Outcomes

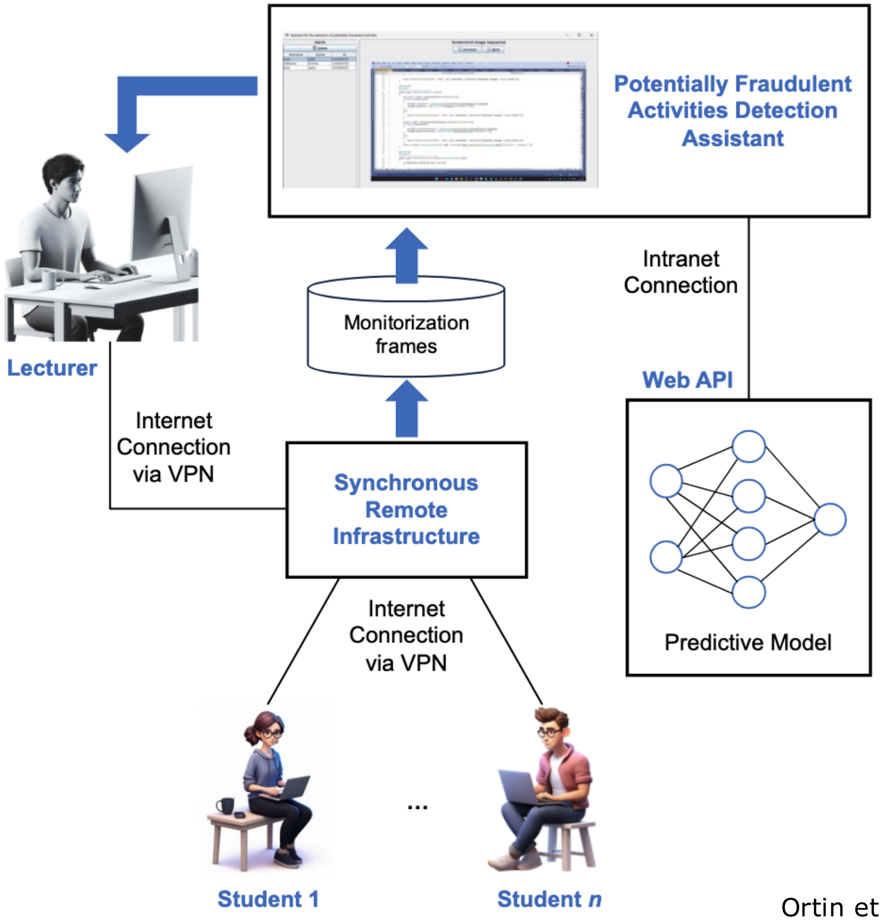
AI Detection Tools

- Tools like TurnItIn can detect plagiarism
- Schools can have their own cheat-detection systems
- AI-generated content tend to have similar styles and patterns
- Generally cheapest and easiest to scale, but can make mistakes

Photo credit: @bruce.digital

Monitored Online Classrooms

- Use monitored online platforms that are monitored in real time
- Studied by (Ortin et al., 2025)
  - Uses intermittent screenshots to detect suspicious activity
  - Achieves 95% accuracy, and an F_2 score of 94.2%



Ortin et al., 2025

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Monitored Online Classrooms

Benefits

- More accurate than traditional cheating detection
- Scalable

Drawbacks

- Requires human oversight
- Breach of privacy

Collaboration and AI Usage Guidelines

- Expectations and limitations should be made clear in course outline

Use of Generative Artificial Intelligence:
The following statement is prepared by the Office of Academic Integrity with input from the Centre for Teaching Excellence, Library, and consultations with Associate Deans and members of the Standing Committee on New Technologies, Pedagogy, and Academic Integrity (Last Updated: August 2023):

Generative artificial intelligence (GenAI) trained using large language models (LLM) or other methods to produce text, images, music, or code, like Chat GPT, DALL-E, or GitHub CoPilot, may be used for assignments in this class with proper documentation, citation, and acknowledgement. Recommendations for how to cite GenAI in student work at the University of Waterloo may be found through the Library.

Please be aware that generative AI is known to falsify references to other work and may fabricate facts and inaccurately express ideas. GenAI generates content based on the input of other human authors and may therefore contain inaccuracies or reflect biases. In addition, you should be aware that the legal/copyright status of generative AI inputs and outputs is unclear. Exercise caution when using large portions of content from AI sources, especially images. More information is available from the Copyright Advisory Committee.

A Discussion on AI Detection...

- Do not disallow AI use without having prevention measures
  - “Honour” systems tend to benefit dishonest students
  - Depends on environment. Take advice with a grain of salt
- Detection tools are a temporary solution
  - As AI improves, this becomes a game of cat-and-mouse
  - Too much prevention erodes student trust

Approaches to Consider

Cheating Detection and Prevention

Alternative Assessment Formats

Re-alignment of Course Outcomes

In-Person Labs and Activities

- Do major assessments in-person
- Dedicate computer lab time to do programming assignments
  - Can allocate TA time to monitor activity
- Have in-lecture activities

In-Person Labs

Benefits

- Easy to monitor
- Students have immediate access to help
- *More efficient allocation of TA resources*

Drawbacks

- More resources needed
- Less time for more complex tasks

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In-Person Labs – Allocation of TA Resources

There has been a trend of decreasing attendance

Less than one student shows up per hour of office hours

On average, less than ten students attend tutorials

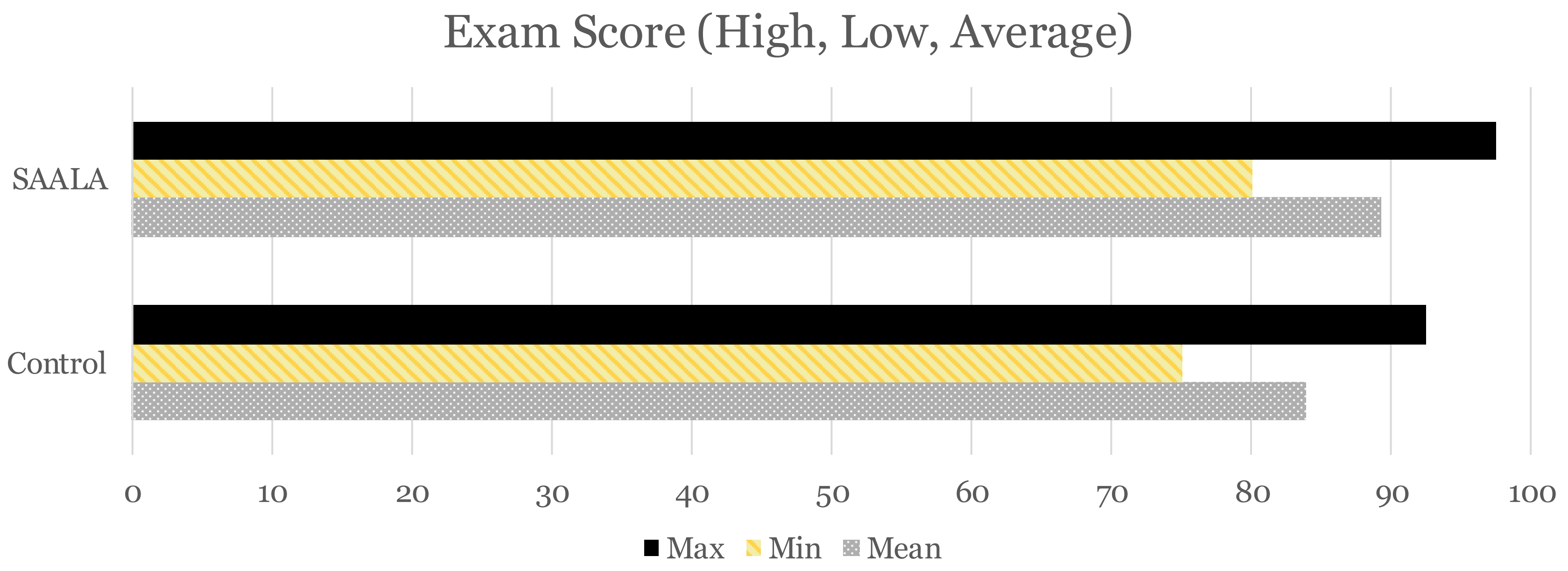
More students are willing to show up to graded activities

5% participation on tutorials resulted in a 60% attendance rate

1-3 students tend to stay around to ask questions at the end of lectures and tutorials

Integration of AI Use In Assessments

- Structured AI-Assisted Learning Approach (Abdullah & Al Musawi, 2026)
  - A structured approach to instruct students on how to use AI effectively
  - Students using SAALA scored higher on closed-book exam



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Approaches to Consider

Cheating Detection and Prevention

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AUTHENTIC ASSESSMENT

"Assessment that is conducted through 'real world' tasks, requiring students to demonstrate their knowledge and skills in meaningful contexts."

(SWAFFIELD, 2011)

Authentic Assessment

- Assessment design should prepare students for a rapidly evolving world (Su et al., 2026, paraphrased)

Scenario – Database Design Process Report

Assignment instructions fed to ChatGPT → ChatGPT Output returned to students → Students tasked to complete assignment using output as baseline

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Technology in Learning - A Basic Example

What did this replace?

What learning goals remained?

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What skills or tools did the calculator replace? What did it not replace?

What skills or tools did the calculator replace? What did it not replace?

RESULTS SLIDE

Technology in Learning - A Basic Example

Replaced:

- Log tables
- Trig tables

Did not replace:

- Mental arithmetic
- Basic numeracy

Enhanced:

- Higher-level problems

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Moving Forwards

What skills are safe to be replaced by AI?

What skills should remain despite being replaceable by AI?

What higher-level skills can be learned with the help of AI?

Summary

- Cheating detection and prevention tools can deter students in the *short term*
- Availability of instructors (e.g. in-person activities) can reduce the need to use AI
  - As a side effect, more efficient use of time
- Ultimately, instructors need to adapt their courses alongside new available technology

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